



Interpreting Logit Coefficients Transcript

Hello everyone, welcome to lesson 1 video 3. In this video, we will learn how to interpret coefficients from logit models. Understanding these or these coefficients mean that what and how crucial they are in practical sense. In this video, let's discuss about from where we left in the last video, right? We understood we can use logistic function, but logistic function gives us values in between 0 and 1. That's what also we know.

Now, what do these numbers mean? What do these numbers in between 0 and 1 mean, okay? So if you think of, you have a data, input data and you could have, you could use a linear combination of your input data or let's say x values. Let's say given your data as age, what is the possibility that the person would default given their age? So you have a very simple model. Let's say you have only one variable, you have very simple model, you have the age of the individual and using the age of the individual, you would want to predict whether the individual is going to default or for the loan that they have taken, okay? Now, you are fitting this model in the data that you have got.

Some of the individuals in your historical data would have actually defaulted, some of them might not have defaulted. In total, you have this entire data and you are using a very simple linear function to relate to that data, okay? Now because your y variable is 0 or 1, basically someone who is likely to default as 1, someone who is not likely to default as 0, so you have the entire historical data in which you have who has defaulted, who has not defaulted and the corresponding age values as well. So you have two columns, one the default value, second the age value.

Now, in this simple situation, let's say, you have to find out the parameters and these parameters β_0 and β_1 would essentially tell you how that model would assume the values. Now, you already have got this, okay? Now think of a logistic model. You have a linear model here, okay? You would have to transform this into a nonlinear model which is an S shaped curve.

That's what we learnt in the last video. But in a logit model, the relationship is nonlinear, okay? So we'll have to actually make it linear in terms of or in some sense. So that's when the log of things would come in.

But essentially what I am trying to say is, we cannot directly say β_1 is the change in probability just like how we said it in the linear regression case, okay? Linear regression case, we said that β_1 is nothing but the rate of change of Y value. Herein we cannot say that β_1 is the rate of change of Y with respect to X . We will not be able to make that assumption because we also have the logit function. So the interpretation in logit model is very different.

So let's say, for making that interpretation, the very first thing that we'll have to do is to define the odds. On the left hand side, you see that odds is the probability of getting the success divided by probability of failure. Probability of success is let's say

probability of Y is equal to S or 1 minus probability of Y is equal to S. This is the probability of failure.

So odds is, think of odds as the ratio of probability of success to probability of failure. Now log of odds can also be written as taking the log of this and you denote it as log of odds. The logit model in that case would assume this shape of log of probabilities of success divided by probability of failure which is the log of odds is equal to β_0 plus $\beta_1 X$. Now on the left hand side, you have log of odds.

On the right hand side, you have the linear combination of input data, right? Let's say the age value here. But if you remember about the logit function, let me show it to you. If you see here, right, if you do a simple maths, let's say if you have to write e to the power β_0 plus $\beta_1 X$ in terms of P, so you can actually do that by taking a simple conjugate and doing simple maths and then finding out what exactly is the value of e to the power β_0 plus $\beta_1 X$. Now that value of e to the power β_0 plus $\beta_1 X$ is nothing but what you actually would be looking here or what you actually are looking here.

So if you think of, if I rewrite it as P by 1 minus P or log of odds, sorry, or the ratio of odds is equals to, it can be rewritten as e to the power of β_0 plus $\beta_1 X$. That's exactly the same thing, right, isn't it? So you can take the exponential of this entire equation and it would get transformed to the odds which is P by 1 minus P is equals to e to the power β_0 plus $\beta_1 X$. That is what I was telling you about here in this case. In the last video, I was discussing about this logistic function, right. So that's how we have deduced this entire equation.

So what does this mean? It means that the log of odds is equals to the linear combination of input data that you have. Now the right hand side of the equation is linear equation, right, and you can easily relate if you take derivative of log of odds with respect to X, you will get β_1 because β_0 would become 0, it is free of X, right, so you will only get β_1 . So it means that β_1 is nothing but the rate of change of log of odds and not the rate of change of probability value itself, okay. So that's the entire intuition I was trying to give you or the entire interpretation that I was trying to give you.

So a one unit change in X is associated with the change in log of odds and not the probability, okay. So this is the interpretation and this is more mathematically correct interpretation and more intuitive as well and we have got a means of using or deriving the entire model as log of odds and connecting it to linear combination of our parameter and the variables that we have in the model, which is the right hand side is a linear model, hence it is also called as linear classifier because we are trying to classify in either 0 or 1, yes or no, true or false and so on, that's why the model itself is called as linear classifier. Now we have moved from log of odds to log of odds ratio, if you remember log of p by 1 minus p is nothing but log of, not log of odds but log of odds ratio, okay.



In the next video, we will discuss more about what is the meaning of odds ratio and how does it make sense. So let's meet in the next video and let's cover most of the rest of the parts that we have in logistic regression. In summary, coefficients of logistic models or logistic regression models helps us understand how log of odds are important in terms of our output variable, which are binary

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